PCA and Dimensionality Reduction

ANSWER KEY

Section 1: Data Centering and the Covariance Matrix

1 Why must data features be centered (subtracting feature means) prior to performing Principal Component Analysis?

a. So the first principal component aligns with the axis of maximum variance rather than the offset from origin ✓

b. To make the data non-linear

c. To ensure all eigenvalues are equal to 1

d. To reduce the matrix dimension

Without centering, the first component simply points toward the data mean rather than capturing internal variation.

2 For a zero-mean data matrix $\mathbf{X} \in \mathbf{R}^{N \times D}$, state the sample covariance matrix $\mathbf{\Sigma}$.

a. $\mathbf{\Sigma} = \frac{1}{N-1}\mathbf{X}^T\mathbf{X}$ ✓

b. $\mathbf{\Sigma} = \frac{1}{D-1}\mathbf{X}\mathbf{X}^T$

c. $\mathbf{\Sigma} = \mathbf{X}^T\mathbf{X}$

d. $\mathbf{\Sigma} = \frac{1}{N}\mathbf{X}^2$

Each element $\Sigma_{ij} = \frac{1}{N-1}\sum_k X_{ki}X_{kj}$, which in matrix form is $\frac{1}{N-1}\mathbf{X}^T\mathbf{X}$.

3 What is the mathematical symmetry property of the covariance matrix $\mathbf{\Sigma}$?

a. It is symmetric and positive semi-definite (all eigenvalues $\lambda_i \geq 0$) ✓

b. It is skew-symmetric

c. It is an orthogonal matrix

d. It has negative trace

$\mathbf{\Sigma}^T = \left(\frac{1}{N-1}\mathbf{X}^T\mathbf{X}\right)^T = \frac{1}{N-1}\mathbf{X}^T\mathbf{X} = \mathbf{\Sigma}$, and variance is always non-negative.

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Section 2: Eigenvalues, Eigenvectors and Variance Maximization

4 A 2×2 covariance matrix has entries $\Sigma_{11} = 4$, $\Sigma_{22} = 1$, and covariance $\Sigma_{12} = \Sigma_{21} = 2$. Find its eigenvalues.

a. $\lambda_1 = 5, \quad \lambda_2 = 0$ ✓

b. $\lambda_1 = 4, \quad \lambda_2 = 1$

c. $\lambda_1 = 3, \quad \lambda_2 = 2$

d. $\lambda_1 = 2, \quad \lambda_2 = 2$

$\det(\mathbf{\Sigma} - \lambda \mathbf{I}) = (4-\lambda)(1-\lambda) - 4 = \lambda^2 - 5\lambda = \lambda(\lambda - 5) = 0 \Rightarrow \lambda = 5, 0.$

5 What is the geometric meaning of the first eigenvector $\mathbf{v}_1$ corresponding to largest eigenvalue λ_1?

a. **The direction along which the data exhibits maximum variance** ✓

b. The direction perpendicular to all data points

c. The direction with the lowest information content

d. The vector of class centroids

PCA seeks $\max \mathbf{v}^T \mathbf{\Sigma} \mathbf{v}$ subject to $\|\mathbf{v}\| = 1$. The solution is the eigenvector of $\lambda_{\max}$.

6 If $\mathbf{v}$ is an eigenvector of $\mathbf{\Sigma}$ with eigenvalue λ, what is the variance of data projected onto $\mathbf{v}$?

a. λ ✓

b. $\sqrt{\lambda}$

c. λ^2

d. 1

$\text{Var}(\mathbf{X}\mathbf{v}) = \mathbf{v}^T \mathbf{\Sigma} \mathbf{v} = \mathbf{v}^T (\lambda \mathbf{v}) = \lambda \|\mathbf{v}\|^2 = \lambda.$

Section 3: Explained Variance and Scree Analysis

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7 A dataset has eigenvalues $\lambda_1 = 70, \lambda_2 = 20, \lambda_3 = 10$. What percentage of total variance is explained by the first two principal components?

a. 90% ✓

b. 70%

c. 80%

d. 20%

Total variance $= 70 + 20 + 10 = 100$. Explained $= \frac{70+20}{100} = 90\%$.

8 How is an 'elbow' in a scree plot (eigenvalues plotted against component index) used in practice?

a. **To select the optimal cutoff k where adding further components yields diminishing returns** ✓

b. To detect outliers in the target labels

c. To determine the learning rate for gradient descent

d. To test for multicollinearity

The elbow point separates significant signal dimensions from noise dimensions.

9 If a dataset has 100 features and you retain the top 10 principal components, by what percentage is feature dimensionality reduced?

a. 90% ✓

b. 10%

c. 50%

d. 80%

Reduction is from 100 to 10 features: $\frac{100-10}{100} = 90\%$.

Section 4: SVD Connection and Applications

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10 Given the Singular Value Decomposition $\mathbf{X} = \mathbf{USV}^T$, how are the principal component directions $\mathbf{v}_i$ directly obtained?

a. **As the right singular vectors (columns of $\mathbf{V}$)** ✓

b. As the left singular vectors (columns of $\mathbf{U}$)

c. As the diagonal elements of $\mathbf{S}$

d. As the eigenvalues of $\mathbf{U}$

$\mathbf{X}^T\mathbf{X} = \mathbf{VSU}^T\mathbf{USV}^T = \mathbf{VS}^2\mathbf{V}^T$, proving columns of $\mathbf{V}$ are eigenvectors of $\mathbf{X}^T\mathbf{X}$.

11 What is the mathematical relation between singular values s_i of $\mathbf{X}$ and eigenvalues λ_i of covariance matrix $\mathbf{\Sigma}$?

a. $\lambda_i = \frac{s_i^2}{N-1}$ ✓

b. $\lambda_i = s_i$

c. $\lambda_i = \sqrt{s_i}$

d. $\lambda_i = s_i(N-1)$

Since $\mathbf{\Sigma} = \frac{1}{N-1}\mathbf{X}^T\mathbf{X}$ and eigenvalues of $\mathbf{X}^T\mathbf{X}$ are s_i^2, $\lambda_i = \frac{s_i^2}{N-1}$.

12 Name a major practical application of PCA in image processing and computer vision.

a. **Eigenfaces: facial compression and feature representation in lower-dimensional eigenspace** ✓

b. Image sharpening through edge convolution

c. Color palette inversion

d. Bilinear image interpolation

Eigenfaces (Turk and Pentland) represents high-resolution faces as linear combinations of dominant eigenvectors.

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